

Ebin Biju

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Professional Summary

Software Engineering Intern and Computer Science undergraduate experienced in Java/Spring Boot and TypeScript/Next.js within a regulated enterprise environment. Builds reliable developer tooling, test automation, secure integrations, and operational diagnostics for distributed event-driven systems.

Professional Experience

American Express

2026 – Present

Software Engineering Intern – Enterprise Platform Engineering

- **Introduced mutation testing** for a Java/Spring Boot service using PiTest, evaluating 450+ mutations across JUnit and Cucumber test suites and establishing an 87% mutation-coverage baseline.
- **Engineered a reusable shared GitHub Actions workflow** for multi-module Gradle services, incorporating validation, automated reporting, and incremental analysis to reduce repeat CI execution time by 80%+.
- **Created a separate reusable shared CI workflow for Maven repositories**, applying the mutation-testing approach to another enterprise Java service and expanding automated test-quality analysis across build systems.
- **Built a full-stack Kafka diagnostics feature** using TypeScript and Next.js with authenticated server APIs, platform-aware configuration, token caching, and group/topic search, validated by 128 automated tests.
- **Investigated distributed E2E test failures** across a 37-scenario suite, strengthening Kafka isolation, asynchronous polling, record correlation, and CI diagnostics.
- **Reduced false-positive operational alerts** in data-processing workflows through configurable rules, integration tests, and support runbooks, while contributing to cloud-migration diagnostics for an internal developer portal.

Innovation & Recognition

1st Place – American Express UK Internal Hackathon: Co-developed and presented the winning concept in a three-person team, subsequently presenting to senior leadership and progressing the concept through the patent-filing process.

Selected Projects

Studio Portal | React, TypeScript, Supabase, PostgreSQL, Vite

- Built a responsive photography-studio platform with role-aware authentication and Row Level Security policies for secure client/admin access.
- Implemented end-to-end enquiry workflows, enabling clients to view and cancel enquiries while administrators manage booking status.

Mortgage Calculator | React, Tailwind CSS, Recharts, Azure, GitHub Actions

- Developed an interactive calculator for 30-year mortgage scenarios with real-time calculations and data visualisation, deployed to Azure through automated CI/CD.

Secure Web Application | Flask, MySQL, Python

- Built an account-management application using bcrypt password hashing, input validation, and protections aligned with OWASP guidance.

Technical Skills

Languages: Java, TypeScript/Javascript, Python, SQL

Backend & Data: Spring Boot, REST APIs, Kafka, PostgreSQL, Supabase, Redis, MySQL

Frontend: React, Next.js, HTML, CSS, Tailwind CSS, Vite

Testing & Delivery: JUnit 5, Cucumber, PiTest, JaCoCo, Testcontainers, GitHub Actions, Docker, Azure, Git

Education

Keele University

Expected 2027

BSc (Hons) Computer Science

- **First-year average: 74%**